

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Probability and Statistics

Semester: Fall

Year : 2020
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- 1) From the following distribution of 500 students of a college find the marks if only 20% students had failed and also minimum marks obtained by the top 25% of the students. 7

Marks	0-20	20-40	40-50	50-60	60-80	80-100
No. of students	50	100	150	90	60	50

Also represent the data by histogram so locate the mode.

- 2) The nine measurements that follow are furnace temperatures recorded on successive batches in a semiconductor manufacturing process (units are): 953, 950, 948, 955, 951, 949, 957, 954, 955. 8

Calculate the sample mean, sample variance, and standard deviation.

Construct a box plot of the data.

- 3) The contents of urns I, II and III as follows: 7

1 white, 2 black and 3 red balls.

2 white, 1 black and 1 red balls.

4 white, 5 black and 3 red balls

One urn is chosen at random and two balls drawn. They happen to be white and red. What is the probability that they came from urn II.

- 4) A company produces machine components which pass through an automatic testing machine. 5% of the components entering the testing machine are defective. However, the machine is not entirely reliable. If a component is defective there is 4% probability that it will not be rejected. If a component is not defective there is 7% probability that it will be rejected. 8

- What fraction of all the components are rejected?

- 5) What fraction of the components rejected are actually not defective?

iii) What fraction of those not rejected are defective?

3. a) let X be a continuous random variable with p.d.f.

$$P(x) = \begin{cases} ax & ; 0 \leq x \leq 2 \\ = a & ; 2 \leq x \leq 4 \\ = ax+3a & ; 4 \leq x \leq 6 \\ \text{Otherwise} & \text{elsewhere} \end{cases}$$

- Determine the constant 'a'.
- find $p(X \leq 1.5)$ and $p(1.5 \leq X \leq 2.5)$
- Obtain $E(X)$.

b) Joint probability distribution of X and Y is given by;

$$f(x,y) = 4xy e^{-(x^2+y^2)} : x > 0, y > 0$$

- Test whether X and Y are independent or not?
- Find the conditional density of X given $Y=y$.

a) The number of accident in a year attributed to taxi driver in a city is poisson distribution with mean 3. Out of 1000 taxi driver, find approximately the number of taxi drivers with i) more than 3 accidents in a year) less than 2 accidents in a year.

b) In an examination, 25% of the student got marks more than 80 and 30% of the students got marks less than 45. Assuming the marks to be normally distributed, find mean and standard deviation of the marks of the students.

a) What do you mean by unbiased estimator? Consider a population of 5 units with 4, 2, 10, 6 and 8. Show that sample mean of size 2 is unbiased estimator of population mean.

b) The quantity demanded of a certain product before advertisement and after advertisement is given below:

Before	50	54	60	58	52	48	53
After	52	56	58	62	55	52	53

Is advertisement effective? Test at 5% Level of significance.

a) A manufacture of gun powder has developed a new powder which is designed to produce a muzzle velocity equal to 3000 ft/sec. seven shells are loaded with the charge and muzzle velocities measured. The resulting velocities are as follows: 3005, 2935, 2965, 2905, 2995 and 3905. Does these data present sufficient evidence to indicate that the average velocity differs from 3,000 ft/sec?

The following tables gives the ages and blood pressure of 10 women.

(X)	56	42	36	47	49	42	60	72	63	55
and sure(Y)	147	125	118	128	145	140	155	160	149	150

8

- Find the coefficient of determination and interpret it.
- Find the regression equation Y on X.
- Find the standard error.
- Estimate the blood pressure of a women whose age is 45 years.

short notes on: (Any two)

2x5

Applications of statistics in engineering

Binomial distribution and condition to apply

Sampling and estimation with criteria of good estimator

2020 Fall- P4S

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Attempt all the questions.

- a) From the following frequency distribution.

Obtained Marks	10-20	20-30	30-40	40-50	50-60	60-70	70-80
No. Of students:	7	12	19	26	11	8	6

Construct an ogive that will help you the answer to find the number of students securing marks.

- Less than 45 marks
 - Between 30 and 70 marks.
 - More than 35 marks.
- b) The life of two model A and B of polythene bags in a recent survey shown below.

Bursting pressure (in lb.)	No. of Polythene bags	
	A	B
5-10	2	9
10-15	9	11
15-20	29	18
20-25	44	32
25-30	11	17
30-35	5	13

- What is the average life of each model?
- Which model has greater uniformity and why?

2. a) From a group of 4 engineers, 2 physicists and 3 economists, 3 persons are selected at randomly. Find the probability that:
- Consist of each kind.
 - 1 engineer and 2 economists.
 - At least one physicist.

- b) State Baye's theorem and discuss its applications. The probability that the person has COVID is 0.03. PCR tests are available to determine whether the person actually has COVID. If COVID actually present, the probability that the PCR test will give a positive result (indicating that disease is present) is 0.09. If COVID is not actually present, the probability of a positive test result (indicating that it is present) is 0.02. What is the probability that COVID is actually present if a positive result has occurred (indicating the disease is present)?

3. a) A Random variable X had the following probability function.

X	-2	-1	0	1	2
P(X)	0.2	K	0.4	2k	k

- Find the value of K
 - $E(X)$, $E(4X+5)$, $\text{Var.}(X)$ and $\text{Var}(4X+5)$
- b) An experiment succeeds twice as often as it fails. Find the chance that in the next six trials there will be at least 4 successes. (Binomial distribution)
4. a) In an examination 15% of the candidates got first class (60% marks or above), while 40% failed (securing below 40% marks). Assuming the marks to be normally distributed, estimate the mean and standard deviation.
- b) If two random variables have the joint probability density function.

$$F(x,y)=\frac{2}{3}(x+2y); 0 < x < 1, 0 < y < 1.$$

- Find marginal density function of X & Y.
 - Find the conditional density of X given that $Y=y$.
 - Are X and Y independent.
5. a) A random sample of size 64 has been drawn from a population with standard deviation 20. The mean of the sample is 80. Calculate 95% confidence limits for the population mean. How does the width of the confidence interval change if sample size is 256 instead?

- b) 1000 articles from factory A are examined and found 97% of good quality, 1500 similar articles from factory B are found only 98% of good quality. Would you conclude that products of factory B are superior to those of factory A. Test the hypothesis at $\alpha = 5\%$.

- a) An I.Q. test was administrated to 5 persons before and after they were trained. The results are given below: 7

Candidates:	I	II	III	IV	V
IQ before training:	110	120	123	132	125
IQ After training:	120	118	125	136	121

Test at 5% level of significance that the training was success.

- b) A chemical company wishing to study the effect of extraction time on the efficiency of an extraction operation obtained the data as follows: 8

Extraction time in minute (x): 27, 45, 41, 19, 35, 39, 19

Extraction efficiency in %(y): 57, 64, 80, 46, 62, 72, 52

- Find the regression equation of y on x and estimate the y if $x=40$
- Find the coefficient of correlation between x and y
- Determine coefficient of determination and interpret it.
- Compute standard error of estimate.

7. Write short notes on: (Any Two) 2x5

- Error in hypothesis testing
- Probability and non-Probability Sampling
- Application of statistics in engineering

pqs-2021 F

POKHARA UNIVERSITY

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Attempt all the questions.

- a) The following are the marks obtained by two students A and B in 10 tests 100 marks in each: 7

Test	1	2	3	4	5	6	7	8	9	10
Marks by A	44	80	76	77	65	72	68	56	90	80
Marks by B	48	75	54	60	63	69	72	51	57	66

Determine i) who is better student? ii) who is more consistent?

- b) A random sample was taken of the thickness of insulation in transformer windings, and the following thickness in mm were recorded: 8

18	21	22	29	25	31	37	38	41	39
44	48	54	56	56	57	47	38	35	36
29	37	32	42	43	40	48	36	37	37

- i) Prepare a stem and leaf display
ii) Prepare a box and whisker plot and interpret the shape of the distribution.
- a) A bag contains 10 balls, 2 of which are red, 3 blue and five black. Three balls are drawn at random from the bag. What is the probability that? 8
- i) The three balls are of different colours
ii) Only two balls are of the same colours
iii) The balls are all of same colour?
- b) The chances of X, Y, Z becoming manager of a certain company are 4:2:3. The probabilities that bonus scheme will be introduced if X, Y, Z becomes manager are 0.3, 0.5 and 0.8 respectively. If the bonus scheme has been introduced, what is the probability that : 7

- i) X is appointed as the manager?
- ii) Y is appointed as the manager?

3. a) Let X be continuous random variable with probability density function.

$$f(x) = \begin{cases} ax, & 0 \leq x \leq 2 \\ a, & 2 \leq x \leq 4 \\ ax + 3a, & 4 \leq x \leq 6 \\ 0, & \text{otherwise} \end{cases}$$

- i) Find the value of 'a'.
- ii) Find $P(X \leq 1.5)$ and $P(1.5 \leq X \leq 2.5)$

b) Test for impurities commonly found drinking water from private wells showed that 30% of all wells in a particular have impurity A. if a random sample of 5 wells is selected from the large number of wells in the country, what is the probability that:

- i. Exactly 3 wells have impurity A?
- ii. At least 3 wells have impurity A?
- iii. Fewer than 3 wells have impurity A?

4. a) As a result of tests on 20,000 electric bulbs manufactured by a company it was found that the life time of the bulb was normally distributed with an average of 2040 hours and the standard deviation of 60 hours. On the basis of the information:

- i) Estimate the number of bulbs that are expected to burn for less than 1960 hours.
- ii) Estimate the number of bulbs that are expected to burn in between 1860 hours to 2120-hours.
- iii) Find the minimum lifetime of 2,000 bulbs which have greater lifetime

b) Suppose X and Y have the following joint probability mass function

X \ Y	2	3
1	0.1	0.15
2	0.2	0.3
3	0.1	0.15

Find

- i) The marginal distribution of X and Y
- ii) The conditional probability of Y given that $X=3$
- iii) $F(2,2)$ and $F(3,2)$
- iv) Variance of X and variance of Y

a) A random sample of 100 articles selected from a hatch of 2000 article shows that the average diameter of the article is 0.354 with standard deviation 0.048. Find the 95% confidence interval for the average of batch of 2000 articles. 8

b) A manufacture of gun powder has developed a new powder which is designed to produce a muzzle velocity equal to 3000 ft/sec. seven shells are loaded with the charge and muzzle velocities measured. The resulting velocities are as follows: 3005, 2935, 2965, 2905, 2995 and 3905. Does these data present sufficient evidence to indicate that the average velocity differs from 3,000 ft/sec? 7

a) For a random sample of 10 pigs fed on diet A the increase in weight (in lbs) in a certain period was 10, 17, 13, 12, 9, 8, 14, 15, 6, and 16. For another random sample of 12 pigs fed on diet B, the increase in weight in the same period were 14, 18, 8, 21, 23, 10, 17, 12, 22, 15, 7, 13. Test whether diets A and B differ significantly as regards their effect on an increase in weight. 7

b) In the United States, during the fall harvest season Pumpkin are sold in the large quantities at the farm stands. Often instead of weighing the Pumpkin prior to sale, the farm stand operators will just place the Pumpkin in the appropriate circular cut out on the counter. When asked why this was done one farmer replied, "I can tell the weight of the pumpkin from its circumference". To determine whether this was really true; a sample of 8 Pumpkin's circumference and weights were measured: 8

Circumference in cm	50	55	54	52	37	52	53	47
Weight in grams (00)	12	20	15	17	9	14	16	14

- Develop the estimating linear regression equation to predict the weight of the pumpkin that is 60 cm in circumference.
- Calculate the Standard error of the estimate and interpret the result.
- What percentage of the total variation in weight of the pumpkin is explained by the circumference of the pumpkin?

2x5

7. Write short notes on: (Any two)

- Error in hypothesis testing
- Gamma and Beta distribution
- Baye's Theorem

p&s-2021 SP

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Attempt all the questions.

1. a) From the following distribution of 500 students of a college, find the marks if only 20% students had failed and also find the minimum marks obtained by the top 25% of the students. 7

Marks	0-20	20-40	40-50	50-60	60-80	80-100
No. of students	50	100	150	90	60	50

Also represent the data by histogram and locate the mode.

- b) The scores of two golfers for 10 rounds each are: 8
- A: 58 59 60 54 65 66 52 75 69 52
- B: 84 56 92 65 86 78 44 54 78 68

Which may be regarded as the more consistent player?

2. a) Define the mathematical definition of probability. From a group of 3 engineers, 4 economists and 5 statisticians, a sub-group of 4 persons are selected at random from the group. Find the probability that the sub-group will consist of 8
- 2 engineers and 2 economists
 - 1 engineer, 1 economist and 2 statistician
 - All statisticians.

- b) Customers who purchase a certain make of car can order an engine in any of three sizes. Of all cars sold, 45% have the smallest engine, 35% have the medium-sized one, and 20% have the largest. Of cars with the smallest engine, 10% fail an emissions test within two years of purchase, while 12% of those with the medium size and 15% of those with the largest engine fail. What is the probability that a randomly chosen car will fail an emissions test within two years? 7

3. a) a random variable X has the following probability function,

$X:$	2	3	4	5	6	7
$F(X):$	0.1	K	0.2	$2K$	0.3	K

Find the value of K and find mean and variance.

- b) In a certain factory turning out optical lenses, there is a small chance $1/500$ for any lens to be defective. The lenses are supplied in a packet of 10. Use Poisson distribution to calculate the approximate number of packets containing no defective, one defective, two defective and three defective lenses respectively in a consignment of 10,000 packets.
4. a) In a class quiz, 8% of the students obtained marks below 25 and 90% of the students got below 85 marks. Assuming that the marks are normally distributed, find the mean and standard deviation of the distribution.
- b) Let X and Y be two continuous random variables with joint probability density function
- $$f(x, y) = Axy; \quad 0 < x < 1 \text{ and } 0 < y < x$$
- Find the constant A .
 - Find the marginal density functions of X and Y .
 - Find the conditional density function of Y given $X = x$.
 - Check the independency of X and Y .
5. a) A random sample of 8 envelopes is taken from letter box of post office and their weights in gms are found to be 12.1, 11.9, 12.3, 11.9, 12.1, 12.4, 12.1, and 11.5. Find 99% fiducial limits for the mean weight of envelopes received at the post office.
- b) What do you mean by p-value? In a factory 2% fans were found to be defectives in a lot of 3000 fans and in other factory 3% fans were found to be defective in a lot of 2500 fans. Do you find that the fans in secondary factory are significantly inferior compared to the fans in the first factory? (Use $\alpha=5\%$)
6. a) Resistors labeled as hundred ohm are purchased from two different vendors. The specification for this type of resistor is that its actual resistance be within 5% of its labeled resistance. In a sample of 180 resistors from vendor A, 150 of them met the specification. In a sample of 270 resistors purchased from vendor B, 233 of them met the specification. Vendor A is the current supplier, but if the data demonstrate convincingly that a greater proportion of the resistors from vendor B meet the specification, a change will be made.
- State null and alternative hypothesis.

- ii. Calculate test statistic and draw a conclusion that a change should be made? (Use $\alpha=5\%$)
- b) The following table presents shear strengths (in kN/mm) and weld diameters (in mm) for a sample of spot welds. 8

Diameter	4.2	4.4	4.6	4.8	5.0	5.2	5.4
Strength	51	54	69	81	75	79	89

- i. Compute the least-squares line for predicting strength from diameter.
- ii. Predict the strength for a diameter of 5.5 mm.

Write short notes on: (Any two) 2x

- Error of hypothesis testing
- Confidence Interval
- Probability Sampling

p2s-2022 Fall

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POKHARA UNIVERSITY
Semester: Spring

Year : 2023
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Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

- a) In a survey, it was found that 64 families bought milk in litre. A frequency distribution is given as 8

Milk in litres	Frequency
5 - 9	7
10 - 14	10
15 - 19	13
20 - 24	18
25 - 29	8
30 - 34	5
35 - 39	3

Construct the histogram and frequency polygon of the distribution.

- b) The average electricity consumption and standard deviation of two sets of households are as follows. 7

	Set I	Set II
Mean	65 units	72 units
Standard deviation	13 units	18 units
Number of households	50	40

- Which set is more variable?
 - Calculate the coefficient of variation of the combined set.
- a) From a group of 3 Nepalese 4 Indians and 5 Americans a committee of 4 persons are selected at randomly. Find the probability that the committee consists 8
- 3 Nepalese and 1 Indian
 - 1 Nepalese, 1 Indian and American

iii. 4 Americans

- b) In a certain factory machines, I, II, and III are all producing springs of the same length of their production machine I, II and III produces 1%, 2%, 3% defective springs respectively. Of the total production the machine I, II and III produces 35%, 25% and 40% respectively. If one spring is selected at random from total springs produced in a day, find
- The probability that it is defective
 - The conditional probability that it was produced by machine III

3. a) Define Mathematical Expectation. An unbiased coin is tossed until a head appears. What is the expectation of number of tosses required to get first head?

b) Define Standard Normal distribution. In a normal distribution 31% of the items are under 45 and 8% are over 64. Find the mean and standard deviation of the distribution

4. a) The number of accidents in a year attributed to taxi driver in a city is Poisson distribution with mean 3. Out of 1000 taxi drivers find the number of taxi drivers

- More than 3 accidents in a year.
- Less than 2 accidents in a year.
- No accidents in a year.

b) Define joint and marginal probability density functions. Two random variables X and Y have the following probability density function.

$$f(x, y) = \begin{cases} 8xy & 0 < x < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Find:

- Marginal probability density functions of x and Y
- Conditional functions of X on y Y on X

5. a) Define Interval estimation. A random sample of 100 articles selected from a batch of 2000 articles shows that the average diameter of the article is 0.354 with standard deviation 0.048. Find 95% confidence limits for the average diameter of this batch of 2000 articles.

b) In a random sample of 400 students of the university teaching departments, it was found that 300 students failed in the examination. In another random sample of 500 students of the affiliated colleges, the number of failures in the same examination was found to be 300. Find

out whether the proportion of failures in the university teaching departments is significantly greater than that of affiliated colleges. Use $\alpha = 0.05$.

6. a) An I.Q. test was administered to 5 persons before and after they were trained. The results given below: 7

Candidates	1	2	3	4	5
IQ before Training	110	120	123	132	125
IQ after training	120	118	125	136	121

Is the training effective at 5% level of significance?

- b) The following are marks obtained by 9 students in Mathematics (X) and Statistics (Y) : 8

X	40	65	60	25	85	35	45	70	80
Y	30	85	65	35	90	35	75	75	45

- Develop the estimating regression equation of X on Y.
- Find the correlation coefficient
- Find Coefficient of determination.

7. Write short notes on: (Any two) 2×5

- Properties of a good Estimator.
- Error in hypothesis testing.
- Methods of primary data collection.

PRS - 2023 Spring